



جامعة الإمام محمد بن سعود الإسلامية
IMAM MOHAMMAD IBN SAUD
ISLAMIC UNIVERSITY

الكلية التطبيقية
APPLIED COLLEGE

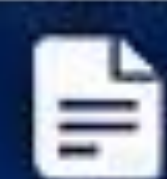
Secure Cloud Sandbox

Cloud-Based Isolated Environment for Threat Analysis

Prepared by:

- Saad Al-Mutairi
- Rayan Almutawa
- Ayed Aldosari
- Abdullah Almughamis

Supervisor:
● Mohammed Falah Alajmi



ABSTRACT

Secure Cloud Sandbox is a cloud-based security system that analyzes suspicious files and links inside isolated virtual machines (VMs). The system monitors behavior, detects malicious activities, and generates security alerts without affecting real environments. It improves cloud security using virtualization and threat analysis technologies.



INTRODUCTION

Cyber threats are increasing rapidly in modern cloud environments. Organizations need secure methods to analyze suspicious content without risking their systems. Secure Cloud Sandbox provides an isolated cloud environment that safely executes and monitors suspicious files using virtualization and cloud security technologies.



OBJECTIVES

- Analyze suspicious files and links in an isolated cloud environment.
- Detect malicious behavior using monitoring and threat analysis.
- Protect systems and networks from potential cyber threats.
- Improve cloud security using virtualization technologies.



TECHNOLOGIES USED



Virtual Machines (VMs) for isolated execution.



Cloud Infrastructure (e.g., AWS / Azure).



Behavior Monitoring using system logs and resource tracking.



Rule-based detection and behavior analysis.



Secure network isolation using virtual network boundaries.



SYSTEM ARCHITECTURE



1. USER UPLOAD

User uploads suspicious files or links to the cloud sandbox.



2. UPLOAD API / GATEWAY

The request is received through a secure cloud API gateway.



3. SANDBOX CONTROLLER

The controller validates the request and prepares an isolated environment.



4. ISOLATED VIRTUAL MACHINE (VM)

The file is executed inside an isolated virtual machine (VM) with no internet access.



5. BEHAVIOR MONITORING

System activity, file behavior, network traffic, and resource usage are monitored in real-time.



6. THREAT DETECTION ENGINE

Abnormal or malicious behavior is detected using rule-based detection and behavior analysis.



7. SECURITY REPORT

A detailed analysis report and security alerts are generated for the user.



Cloud Component



Virtual Environment



Security Component



Output Report



METHODOLOGY

1. Upload suspicious files or links to the cloud sandbox.
2. Execute the files inside isolated virtual machines (VMs).
3. Monitor file behavior and system activity.
4. Detect malicious actions and generate security alerts.
5. Produce a security report with analysis results.



RESULTS



Safe analysis of suspicious files inside isolated virtual machines.



Detection of malware, unauthorized access, and abnormal behavior.



Generation of automated security alerts and detailed reports.



Reduces the risk of system infection by isolating suspicious files.



Faster threat detection and response (alerts in seconds).



Enhanced protection of systems and sensitive data.



CONCLUSION

Secure Cloud Sandbox provides a secure cloud environment for analyzing suspicious files and detecting cyber threats using virtualization, monitoring, and rule-based behavior analysis.



CLOUD ENVIRONMENT



AWS EC2 (Virtual Machines)



AWS S3 / Azure Storage (File Storage)



VPC / Virtual Network (Network Isolation)



CloudWatch / Azure Monitor (Monitoring & Logs)



APPLICATIONS

- Malware analysis
- Suspicious link analysis
- Internal security operations
- Incident response & forensics
- Security testing and validation



KEY BENEFITS

- ✓ Isolated and safe threat analysis
- ✓ Real-time monitoring and alerts
- ✓ Protects cloud resources and data
- ✓ Reduces risk of cyber attacks
- ✓ Improves overall cloud security



SECURE CLOUD SANDBOX
ISOLATE • ANALYZE • PROTECT